United States Patent [19]

Pippin, Jr.

[11] **3,962,818**

[45] **June 15, 1976**

[54] **REACTION TOY ARRANGEMENT AND METHOD**

[76] Inventor: **Reginald F. Pippin, Jr.**, 7806 Ruxway Road, Towson, Md. 21204

[22] Filed: **June 22, 1972**

[21] Appl. No.: **265,283**

Related U.S. Application Data

[62] Division of Ser. No. 102,294, Dec. 29, 1970, Pat. No. 3,683,544, which is a division of Ser. No. 504,300, Oct. 24, 1965, Pat. No. 3,550,313.

[52] **U.S. Cl.** ..... **46/74 D**

[51] **Int. Cl.**[2] **A63H 27/100**

[58] **Field of Search** ..... 46/74 R, 74 C; 244/1, 244/74

[56] **References Cited**

UNITED STATES PATENTS

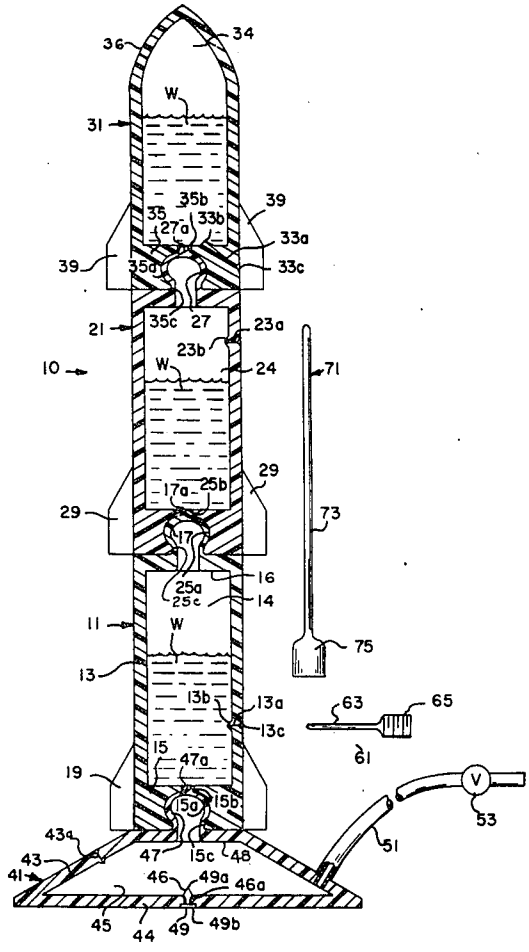
3,046,694	7/1962	Holderer	46/74 C
3,049,832	8/1962	Joffe	46/74 C
3,091,052	5/1963	Ryan	46/74 C
3,174,501	3/1965	De See	273/58
3,550,313	12/1970	Pippin	46/74 R
3,785,083	1/1974	Pippin	46/74 C

Primary Examiner—Louis G. Mancene
Assistant Examiner—Robert F. Cutting

[57] **ABSTRACT**

A fluid discharge reaction-propelled toy missile arrangement capable of formation in multiple stages or units releasably interconnected as a function of internal propulsive pressure in the succeeding stages or units, or capable of individual unit single stage use, if desired. A self-sealing lateral inlet valve separate and spaced from the fluid discharge propulsion orifice of an individual missile unit enables individual unit pressurization from an external fluid pressure source. Missiles are launched from a ground-launching unit which may have a fluid pressurizing inlet and a fluid pressurizable releasable missile-holding connection, or may have other releasable missile-holding connection, and launch simulation enhancement can be provided, if desired, by controlled-release lateral liquid spray from one or more discharge orifices in the launching unit and separate from the fluid pressurizing inlet therefor.

43 Claims, 9 Drawing Figures



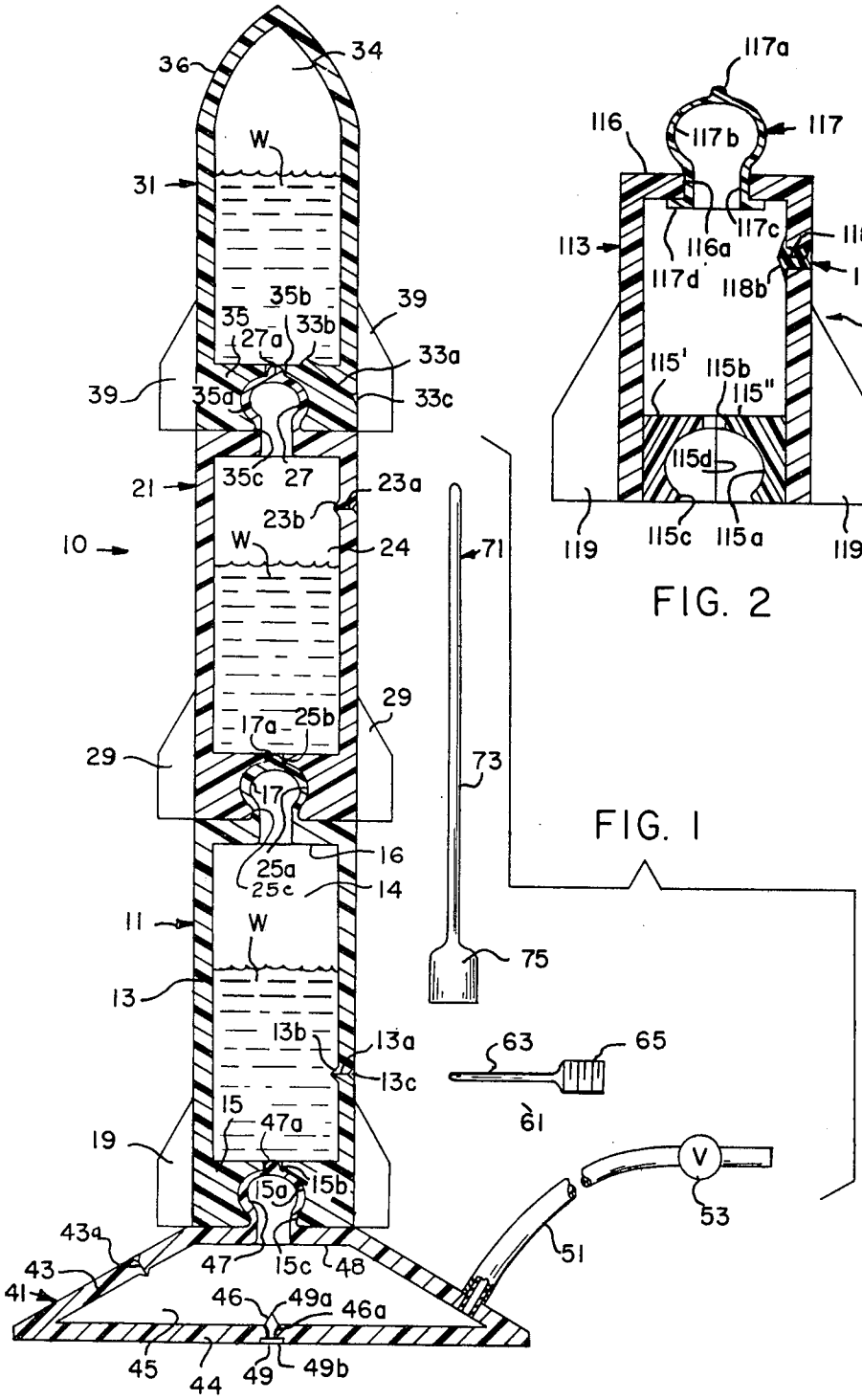

36
34
W
31
35b
39
35
33b
27a
33a
39
35a
33c
21
35c
27
23a
23b
10
W
24
71
29
17a
25b
29
73
17
16
25a
25c
14
11
W
13
13b
13a
13c
15
47a
19
15a
15b
43a
43
47
15c
48
41
46
49a
46a
45
44
49
49b
63
65
61
V
53
51
75

FIG. 1

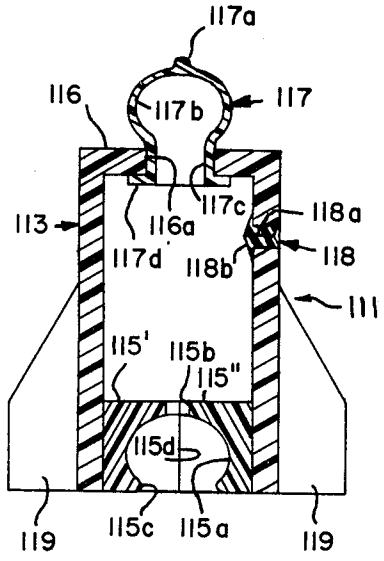

117a
116
117b
117
117c
116a
113
118a
117d
118b
118
111
115'
115b
115"
115d
119
115c
115a
119

FIG. 2

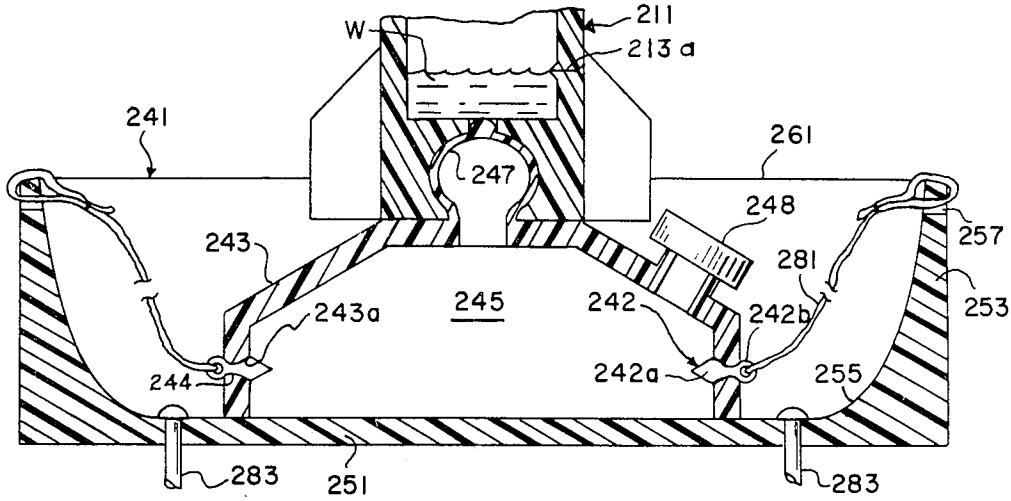


FIG. 3

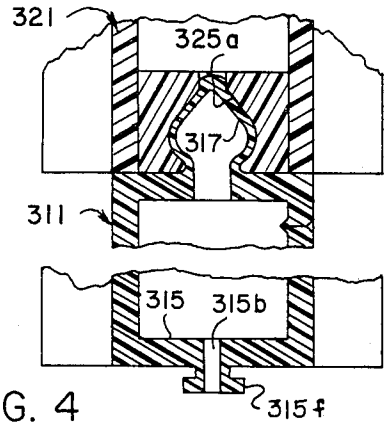


FIG. 4

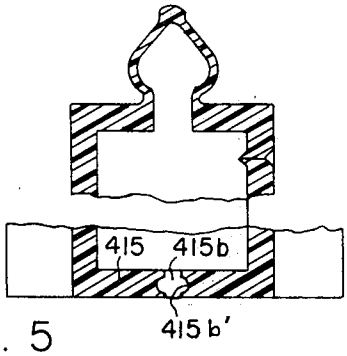


FIG. 5

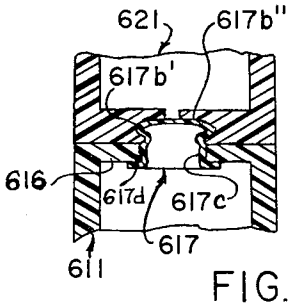


FIG. 9

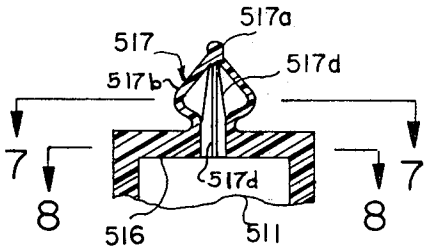


FIG. 6

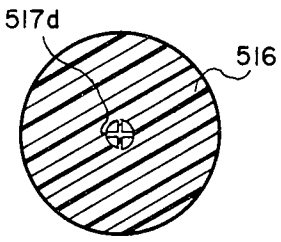


FIG. 8

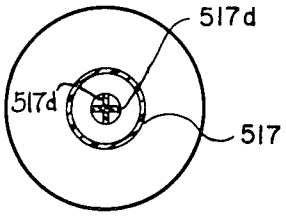


FIG. 7

REACTION TOY ARRANGEMENT AND METHOD

This is a division of co-pending application Ser. No. 102,294, filed Dec. 29, 1970, now U.S. Pat. No. 3,683,544, Aug. 15, 1972, which in turn is a division of application Ser. No. 504,300, filed Oct. 24, 1965, now U.S. Pat. No. 3,550,313, dated Dec. 29, 1970.

This invention relates to a reaction toy arrangement and method, and more particularly to a fluid expulsion type reaction motor missile arrangement.

It is a feature of the invention to provide a fluid pressurizable discharge propulsion missile in which the fluid pressure is applied through a pressurizing valve spaced from the fluid discharge propulsion orifice of the missile, the pressurizing valve being preferably self-closing and disposed in a side wall of the missile.

Still a further feature is the provision of a multistage-toy interconnection arrangement in which the various stages are hermetically sealed from one another.

Another feature is the provision of an externally fluid pressurizable discharge propulsion toy missile unit, launchable singly or in multiple stage combination, in which fluid pressurization is accomplished from an external self-sealing valve connecting between the exterior of the missile unit and an internal fluid chamber which also is in open fluid connection to the fluid discharge propulsion orifice of the missile unit.

A further feature is the provision of a toy missile arrangement having multiple fluid pressurizable units with releasably closable respective fluid discharge orifices, and at least one of the units being pressurizable from a source of fluid under pressure through an inlet passageway separate from its fluid discharge orifice.

Another feature is the provision of a toy missile arrangement having multiple units, at least one of which units is internally fluid pressurizable for self-propulsion and has a fluid inlet passageway separate from a fluid discharge propulsion orifice therein, for passage of fluid under pressure into said one unit preparatory to self-propulsion by discharge of fluid under pressure from said discharge propulsion orifice.

Another feature is the provision of a toy missile arrangement having a self-propelled unit which is propelled by fluid discharge from a discharge propulsion orifice therein, which unit has a fluid inlet passageway for fluid pressurization of such unit, and which fluid inlet passageway is separate from and additional to the fluid discharge propulsion orifice of the unit.

Still a further feature is the provision of a facile and a desirable method of pressurizing from an external pressure source, a fluid-discharge-propellable missile unit or units having a discharge propulsion orifice, and by which method the externally derived pressurization is accomplished through an opening, preferably self-closing, other than the discharge propulsion orifice of a given missile unit.

Still other objects, features and attendant advantages will become apparent to one skilled in the art from a reading of the following detailed description of several physical embodiments constructed in accordance with the invention, taken in conjunction with the accompanying drawings wherein:

FIG. 1 is a longitudinal section view of a multistage rocket toy according to the invention;

FIG. 2 is a longitudinal section view of a rocket stage modified for ease of molding manufacture;

FIG. 3 is a fragmentary view of a modified launcher and rocket arrrangement;

FIG. 4 is a longitudinal section view of a further modified first stage rocket section;

FIG. 5 is a longitudinal section view of an additional modified first stage rocket section;

FIG. 6 is a fragmentary longitudinal section view of a modified fluid-pressure-responsive bulb connector-seal arrangement;

FIG. 7 is a section view taken on line 7—7 of FIG. 6;

FIG. 8 is a section view taken on line 8—8 of FIG. 6;

FIG. 9 is a fragmentary longitudinal section view of a further connector-seal arrangement modification.

Referring now to the figures of the drawings, in FIG. 1 is shown a multistage rocket toy **10** which in the present embodiment incorporates three stages, although it will become apparent that with the present invention the number of stages is virtually unlimited within the propulsion capabilities of the given rocket size, being for example one, two, three, or more, stages. The illustrative rocket **10** includes three self-separable stages **11, 21, 31,** the lowermost or first stage **11** being releasably mounted on a launching pad **41** having a remote control pressure release launch-control valve **53** connected thereto through a hollow conduit **51.**

Each separable stage or section **11, 21** and **31** includes a cylindrical body having a pressurizable liquid storage chamber **14, 24** and **34** respectively, enclosed by an annular side wall **13, 23, 33** respectively, and front and rear end walls **15, 16; 25, 26;** and **35, 36,** respectively. Stabilizing fins **19, 29, 39** are formed or secured on the outer surface of the respective stage sections **11, 21** and **31,** and the final or upper stage **31** has a tapered or ogive shaped end **36** for decrease of air resistance during flight.

The forward end walls **16** and **26** of the first and second or penultimate stage sections **11** and **21** have a pressure-sensitive laterally flexible diaphragm wall male connector-seal in the illustrative form of a bulbous flexible nose protrusion **17, 27,** respectively, which is as a function of pressure in the respective liquid storage chamber **14, 24,** pressurizable for rigidizing thereof and depressurizable for ease of flexing and discharge thereof from a respective female connection **25***a*, **25***b*, **25***c;* and **35***a*, **35***b*, **35***c* in the adjoining next forward stage **21** and **31** respectively. Each bulbous nose protrusion **17, 27** includes a unitary relatively thin-walled integral diaphragm-like body structure having an intermediate enlarged annulus portion disposed between a smaller tip protrusion and a smaller annular portion adjacent the root end. The bulbous nose protrusions serve as pressure-responsive connector seals which lockingly engage and seal the respective female connecting and fluid discharge or expulsion openings **25***a*, **25***b*, **25***c;* and **35***a*, **35***b*, **35***c*. In the illustrative embodiment, each female connecting and fluid discharge or expulsion opening includes an intermediate annular groove portion **25***a*, **35***a* disposed between a relatively smaller diameter liquid discharge metering orifice **25***b*, **35***b* and a reduced diameter smoothly rounded rim portion **25***c*, **35***c*. The flexible bulb connector-seals **17, 27** are preferably shaped to be generally complementary to the respective mating female openings, although it will be appreciated that differences in shape and size may be accommodated to a degree dependent upon the elasticity of the material forming the bulb-connector seals, in which event when the seal integrity and connection strength are effec-

tively maintained the effective pressurized joint is therefore formed of effectively substantially complementary relation. Some degree of pressure-responsive distension is present in all materials, and the extent of distension employed, if any, will depend upon the extent of manufacturing or design mismating and the elastic distension capabilities of the material forming the bulb protrusions.

While the term "bulbous" or "bulb" as applied to pressurizable laterally flexible wall connector elements **17, 27,** etc., describes in one restricted sense the specific shape of a male shaped connector having an enlarged mid-section, its primary meaning and scope as used herein lies in and encompasses the generic structural and action sense and meaning of a male shaped connector member generally which is capable of lateral swelling, and in each instance having an internal pressure responsive laterally flexible wall section which is pressurizable when in place within a fluid discharge orifice to afford pressure responsively releasable holding action therewith. The fluid-pressure-responsive laterally flexible diaphragm wall connector elements or members **17, 27,** etc., may, of course, be formed with other configurations and constructions than the illustrated and preferred bulbous configurations and constructions.

Launching pad **41** is pressure-responsively connected to the female connecting and fluid discharge openings **15***a*, **15***b*, **15***c* of the first stage section **11** through the medium of a flexible bulb connector-seal **47** similar to interstage flexible bulb connector-seals **17** and **27.**

Although the pressurizing of the various chambers **14, 24, 34** and **45** will tend to properly seat the pre-inserted bulb connector-seals in their respective female openings it may be advantageous and desirable in some instances, particularly in the case of highly flexible bulb connector-seal embodiments to employ a connector bulb insertion aid **71** which may take the form of a rod **73** having a handle **75.** The rod **71** may be inserted through the female fluid discharge opening **15***b*, **25***b* of a particular rocket section to push on and effect manipulation of the respective bulb connection-seal **17** or **27** into its corresponding adjoining next stage complementary female opening **25***a*, **25***b*, **25***c* or **35***a*, **35***b*, **35***c* as the case may be. To enable use of insertion aid **71** with the interconnection between launcher pad **41** and the first stage section **11,** the bottom wall **44** of pad **41** may be provided with a central removable plug **49** aligned with bulb connector-seal **47.** Plug **49** may be formed with a douple tapered nodule detent **49***a* for ease of insertion and removal and a recessed flat head **49***b*.

The rocket stage sections **11, 21** and **31** and a launching pad **41** are each provided with a self-closing valve **13***a*, **23***a*, **33***a* and **43***a*, respectively, to enable pressurizing of the respective chambers **14, 24, 34** and **45,** after assembly of respective adjoining stages, although the launch pad chamber **45** may if desired be pressurized through valve **53** and conduit **51.** Each of the self-closing valves includes pressure-responsive self-obturating inner closure lips **13***b*, **23***b*, **33***b*, and **43***b* respectively and a flared outer guide opening as indicated at **13***c* may also be provided if desired. For purposes of illustration of differing locations for the valves, each of the valves **13***a*, **23***a*, **33***a* is shown at a different location on its respective rocket stage section, although it will be appreciated that for a given assembly all valve openings **13***a*, **23***a* and **33***a* would normally lie in the same general position on the respective different stage sections. A conventional valve needle **61** of the type commonly employed for inflating footballs, basketballs, etc., and a conventional air pump (not shown) may be used in conjunction with valve **13***a*, **23***a*, **33***a*, and **43***a* to effect pressurizing of the respective chambers **14, 24, 34** and **45.**

It will be appreciated that inasmuch as the illustrated first and second stages **11** and **21** are interchangeable, it is a simple matter to reduce the number of stages to two or one by elimination of one or both of stages **11** and **21,** and to increase the number of stages as desired by adding additional stages identical to either stage **11** or **21.**

As an example of assembly and launching of the rocket **10,** water is added to partially fill first stage storage chamber **14** as by pouring into the effective funnel-mouthed opening **15***a*, **15***b*, **15***c* while the section **11** is inverted, and launching pad **41** is thereupon mated with section **11.** While retaining the assembly **11, 41** in inverted position, the chamber **45** is pressurized through valve opening **43***a* or through line **51** to effect secure retaining connection and sealing of the launching pad **41** to the first stage **11** through pressure-responsive bulb and socket connection **47, 15***a*, **15***b*, **15***c*. Thereupon, water is added to partially fill storage chamber **24** in second stage section **21** while holding the second stage in inverted position, and the first stage bulb connector-seal is then inserted into the opening **25***a*, **25***b*, **25***c*, after which the first stage chamber **14** is pressurized through valve opening **13***a* to lock sections **21** to first stage section **11** and the assembly **11, 41.** This same operation is then repeated for partial filling of third stage section **31,** connection thereof to second stage section **21,** and subsequent pressurizing of second stage **21** through valve opening **23***a* to lock sections **21** and **31** together. The assembly operation is then completed by pressurizing final stage chamber **34** through valve opening **33***a*, and the assembly is then ready for launch.

The foregoing method of assembly is most advantageous when the bulb connector-seals **17** et al. are of sufficient rigidity to enable flexing insertion without the aid of tool **71,** as only two pieces need be held together at a time in unsecured connected condition. However when it is necessary or desired to employ the insertion aid tool **71** on all interstage connections, a different assembly procedure may be employed. In this instance the final stage section **31** is first partially filled with water while inverted and thereupon the second stage section **21** is connected thereto by inserting the bulb connector-seal **27** into the opening **35***a*, **35***b*, **35***c*, using the tool **71** through second stage opening **25***a*, **25***b*, **25***c* to aid in this assembly. Water is then added through opening **25***a*, **25***b*, **25***c* to partially fill second stage **21,** and first stage **11** is connected to stage **21** through insertion of bulb connector-seal **17** into opening **25***a*, **25***b*, **25***c*, using tool **71** through opening **15***a*, **15***b*, **15***c*. Water is then added to partially fill the storage chamber **14** of the first stage **11** while continuing to hold the assembly **11, 21, 31** in inverted vertical position, and the launch pad **41** is then connected by removing the plug **49** and manipulating the tool through opening **46** to aid in positioning the bulb connector-seal within the mating connecting and fluid discharge opening **15***a*, **15***b*, **15***c* of the first stage section **11.** Plug **49** is then replaced, and thereupon launching pad chamber **44** is pressurized through either valve opening **43** or conduit **51** and valve **53.** This secures first stage **11** and launch

pad **41** together in sealed relation and chamber **14** is thereupon pressurized through valve opening **13***a* to secure stages **11** and **21** together, whereupon second stage **21** is then pressurized through valve opening **23***a* to lock and seal stages **21** and **31** together, and final stage chamber **34** is then pressurized through valve opening **33***c* to ready the assembly for launching. It will be appreciated that in this last described method of assembly a holding jig or fixture may be used if desired to retain the various sections in position during assembly, although such may be accomplished by hand only if desired, particularly with the aid of a second person.

To launch the rocket **11** from the launch pad **41**, the assembly **41, 11, 21, 31** is turned to a vertically upright position as shown in FIG. **1**, and valve **53** is released to release the pressure within chamber **45.** The pressure withn chamber **14** of first stage **11** will then cause the flexible bulb connector-seal to laterally collapse and be ejected from the opening **15***a*, **15***b*, **15***c* and liftoff will occur with the subsequent continuing discharge of liquid through metering orifice **15***b*. The rocket assembly will continue upward under the reaction force influence of the ejected water mass from orifice **15***b* until the pressure in chamber **14** drops sufficiently to enable the then greater pressure within chamber **24** (which was initially approximately similar to that of adjacent chambers **14** and **34**), to overcome the holding action of bulb connector-seal **27** in opening **25***a*, **25***b*, **25***c*. Upon reaching this pressure difference the second and third section assembly will then separate from the first stage in a manner similar to that of first stage **11** and launch pad **41.** The second and third stage assembly **21, 31** will continue upward together until a similar pressure differential separation occurs between these two stages, and the third stage will thereupon continue upward alone under its own reaction motor thrust power through liquid ejection from metering orifice **35***b*. Upon exhaustion of the liquid from each of the stages to thrust will quickly drop to zero and the stage will thereafter drop to the launch area. Upon recovery of all stages the rocket **11** may be reassembled together with launch pad **41** and launched again.

For ease of manufacturing the various stages and the launcher may take various forms. For instance, as shown in FIG. **2**, illustrating a modified construction of rocket stage section **111**, the body **111** may be formed in several pieces which are then suitably bonded together to form the unitary body. In this modified embodiment the rear end wall is formed separately from the side and forward wall section **113, 116,** as is the self-sealing valve **118** and the bulb connector-seal **117.** The rear end wall is formed as two or more split laterally mating sections **115'**, **115''**, each having a concavo-concave intermediate groove surface **115***a*, samll diameter concavo-convex metering orifice surface **115***b* and reduced diameter neck rim surface **115***c*. With this construction the separately formed pieces may be formed of different materials consistent with their required function. For instance, the bulb connector-seal **117** may be formed of softer more elastic material than body **113,** as may self-sealing valve **118.** Bulb connector-seal **117** is secured to the main body **113** through bonding of an intermediate neck portion **117***c* and flange retaining portion **117***d* to the adjoining area surrounding opening **116** in end wall **116.** Valve **118,** which is preferably of soft pliant rubber, is inserted through an opening in side wall **113** and is bonded thereto and/or held in place by tight friction fit which aids in assuring normally self-sealing of the opening **118***a*. In addition it may be noted that each of the valve openings **118***a*, **18***a*, etc., of the various valves may be formed by puncturing with a sharp small diameter needle, or by other conventionally practiced self-closing valve construction techniques. Further, various other valve constructions may be employed for the valves **118, 18***a*, etc., as desired, it being desirable however that the valves be normally self-closed in the discharge directions.

FIG. **3** illustrates a modified launch pad arrangement which provides added simulated realism to the launching operation. In this modified embodiment the housing **243** for the pressurizable fluid chamber **245** of the launch pad **241** has radially extending liquid discharge orifices **244** formed at circumferentially spaced positions about the annular periphery thereof. The orifices **244** preferably have a flared inner seat end and are releasably closed by substantially complementary pressure releasable plugs **242,** each having a double tapered retention nodule end **242***a* seating on seat **244***a*. The plugs **242** are preferably additionally connected for safety and retention purposes to the launch pad **241** through the medium of a flexible cord or other flexible line **281** secured through spaced eyes **257** formed in the annular outer upstanding rim **253.**

The launch pad **241** may be secured to the ground for stability, if desired, as with anchor pins **283** extending through the base **251** thereof. While not shown, a tool receiving opening and a removable plug may be provided in the bottom **251** of launch pad **241** as in FIG. **1** if so desired, it being preferred in such cases that the removable tool assistance plug be constructed and shaped for substantially more difficult removal than the plugs **242** so as to prevent undesired removal of the plugs during operation of the launch pad **241.**

A female water hose connection is provided in the wall **243** as indicated at **248** for supply of water to and pressurizing of the chamber **245.** Annulus wall **253** has an upwardly concavely curved surface **255** in the path of discharge from the orifices **244** and this surface **255** serves to deflect and disperse water from the orifices **244** upwardly.

In operation of the embodiment of FIG. **3**, a rocket **211** partially filled with water is connected to launch pad **241** through insertion of bulb connector-seal **247** in the associated rear discharge and connection opening of the rocket body. Water is then added to chamber **245** through a hose (not shown) connected to hose connector **248** sufficient to pressurize the bulb **247** to operational sealing and holding condition, but insufficient to effect discharge of said plugs **242** from their orifices **244.** The rocket **211** is then pressurized through its associated valve **213***a*, whereupon the water pressure is increased within chamber **245** by remote control from a conventional water faucet supply connection (not shown) to effect discharge of the plugs **242** and resulting spraying of water through the orifices **244.** The radial jets of water will be deflected and dispersed upwardly in a generally ring-shaped pattern, the continuity of which is dependent to a degree upon the number of spacing of orifices **244.** With a sufficient quantity and size of orifices in comparison to the available water flow rate into chamber **245,** the resulting pressure drop will be sufficient to enable launching of rocket **211.** However, launching can be assured by turning the faucet full open to eject the plugs **242** and effect the desired water spray action, and thereupon

turning off the faucet, as the pressure in the chamber **245** will in the final condition be reduced to zero, assuring release and lift-off of the rocket **211** from launch pad **241.**

In FIGS. **4** and **5** are shown two permissible, though to some degree less advantageous, modifications of first stage sections of a multistage rocket according to the invention. In each instance the stage section incorporates a pressurizable flexible bulb connector-seal **317** for connection of two adjoining first and second stages **311** and **321,** the bulb connector-seal **317** being of tear-drop shape for each of insertion and relatively greater resistance to removal. This facet is considered desirable in various instances for all stage bulb connector-seals, the somewhat less advantageous construction lying in the alternative constructions for connection to the launcher. In FIG. **4** the launcher connection is a flanged nozzle **315***f* formed on rear wall **315** and having a fluid discharge orifice **315***b*. With this construction the rocket **311** may be launched with the aid of a mechanical slide release launcher such as shown on U.S. Pat. No. 2,732,657. The discharge orifice **315***b* in the embodiment of FIG. **5** has a shallow small diameter annular groove **415***b'* formed in the discharge orifice **415***b* and is adapted to be launched with the aid of a nodular stemmed launcher as provided in U.S. Pat. No. 3,046,694. Various other hybrid stage sections may be employed as desired, though such are generally considered less desirable than the arrangement in which all stages and the launch pad are connected and separated in a similar manner.

A modified bulb connector-seal **517** is shown in the rocket section embodiment **511** of FIGS. **6–8,** wherein the bulb wall **517***b* has a tear-drop shape for relative ease of insertion in comparison to removal of the bulb connector-seal from its complementary female opening, and in which the bulb is longitudinally reinforced while providing collapsibly lateral flexibility of the wall **517***b*. This is effected by a longitudinal reinforcing rib unit having ribs **517***d* integrally connecting, or secured to, the tip and root end portions of bulb connector-seal **517** while being in disconnection therefrom in the intermediate enlarged girth flexible annular wall portion **517***b*.

FIG. **9** shows a further modified form of flexible bulb connector-seal **617** having a generally squat shape with a relatively large diameter securing rim annular portion **617***b'* compared to the depth thereof and a relatively flattened flexible outer end surface **617***b''*. The opening in wall **616** of rocket stage section **611** is relatively large so as to enable downwardly and inwardly rolling deflection and release of the retaining rim annulus portion **617***b'* from its complementary retention groove in response to decrease in pressure in the connecting liquid storage chamber of stage section **611.** In this embodiment, as in FIG. **2,** the bulb connector-seal **617** is formed separately from the main body of the associated rocket stage section **611,** and is bonded thereto along retaining flange **617***d* and intermediate reduced diameter portion **617***c*, although in some instances it may be sufficient to bond only along flange **617***d*, in which latter instance the downwardly and inwardly rolling action of the bulb diaphragm connector-seal **617** may extend over a greater distance during stage separation.

While the invention has been illustrated and described with respect to various illustrative embodiments, various modifications and improvements may be made. For instance, the liquid and/or other desired fluid inserted in any one or more of the missile units and/or launcher unit may be inserted through the pressurizing valve for the particular unit, as in the pressurized loading of the launcher unit, in lieu of being loaded through the unit discharge orifice or separate launcher fill line or orifice. In addition, in a further modification or improvement, pressurizable flexible wall female inter-unit connectors may be employed in lieu of the illustrated pressurizable male connectors, in which instance the female connector may envelope and releasably engage in peripherally sealing relation a projecting rear section of a next succeeding missle unit or stage, which projecting section may include the discharge orifice section of such next succeeding unit, and thereby effectively closing such discharge orifice of the next succeeding unit during pressurized holding of such unit by the female connector. Also, the spray launcher may be provided with positive orifice-opening-and-missle-release-effecting means, such as a selectively longitudinally movable common release ring in releasable closure-controlling direct or indirect connection with a plurality of initially closed spray orifices, in lieu of the pressure responsively self-releasable multiple plug spray launcher as illustrated in FIG. **3.** Accordingly, it is to be understood that the invention is not to be limited by the illustrative embodiments, but only by the scope of the appended Claims.

That which is claimed is:

1. A jet-propelled toy comprising

a plurality of individual interconnectable separable units,

one of said units having a pressurizable fluid storage chamber and a directionally stable jet propulsion fluid expulsion orifice in fluid connection between said storage chamber and the atmosphere,

and a normally self-closed, fluid-pressure-containing, fluid passage valve in the body of said one unit at other than said fluid expulsion orifice for fluid passage connection between said pressurizable fluid storage chamber and a zone exterior of said one unit when said normally self-closed valve is opened.

2. A jet-propelled toy comprising

a plurality of individual self-separable stages,

said stages being interconnected, each of said stages having a pressurizable fluid storage chamber and a fluid expulsion orifice in fluid connection between said storage chamber and the atmosphere,

said chambers being in permanent fluid disconnection from one another at the zone of interconnection thereof,

said stages being connected at and in sealing relation to said fluid expulsion orifice of one of said stages,

said one stage having a sealable pressurizing enabling means in the body thereof and disposed at other than said fluid expulsion orifice for pressurizing said one stage.

3. A jet-propelled toy according to claim **2,**

said sealable pressurizing enabling means being a self-sealing valve.

4. A jet-propelled toy according to claim **1,**

two of said units being separably interconnected through a fluid-sealed fluid-pressure-sensitive connection formed or secured on one of said units and connecting with an adjoining other of said stages.

5. A jet-propelled toy according to claim **4,**

said units being at least three in number,

each of said stages being interconnected through a fluid-sealed fluid-pressure-sensitive connection according to claim **4.**

6. A jet propulsion toy comprising

a body having a pressurizable fluid chamber and closable fluid jet propulsion discharge orifice connecting between said chamber and the outside atmosphere,

and a fluid pressurizing valve in said body and spaced from said jet propulsion discharge orifice.

7. A toy according to claim **5,**

said pressurizing valve comprising a soft elastic material having a normally self-closed elastic bore with an obturating self-sealing internal end surface.

8. A toy according to claim **6,**

said body having an elongate shape with said jet propulsion discharge orifice being disposed in one longitudinal end thereof,

said pressurizing valve being disposed with its outer end extending through the longitudinally outer surface of said body.

9. A toy according to claim **8,**

said body having an interstage connector formed by an inflatably flexibly movable wall section adjacent the longitudinal end thereof opposite said jet discharge orifice end.

10. A jet propulsion toy arrangement according to claim **6,**

further comprising a second self-propelled body having a pressurizable fluid chamber and a fluid jet propulsion discharge orifice connecting between said chamber and the outside atmosphere,

said second body being releasably connected to the first said body through fluid pressure responsive connecting and fluid sealing engagement with said propulsion discharge orifice of the first said body.

11. A jet propulsion toy arrangement according to claim **6,** further comprising a second body as set forth in claim **6,**

said second body being releasably connected to said first body through a fluid-pressure-responsive releasable connection.

12. A jet propelled toy arrangement according to claim **11,** further comprising a third self-propelled body as set forth in claim **11,**

said third body being releasably connected to said second body through a fluid-pressure-responsive connection.

13. A multi-unit toy missile arrangement,

at least one unit thereof being individually self-propellable and having a respective pressurizable fluid chamber and associated fluid discharge orifice,

at least one of said units having a body including a pressurizing enabling element as a part thereof and being other than a fluid discharge orifice, for pressurizing said one stage from an external fluid pressure source through insertion of fluid under pressure into said one unit.

14. The method of fuel loading and pressurizing two separable stages of a jet-propelled multistage missile comprising

adding liquid to a liquid chamber within one of said stages,

securing said stages together through a pressure-sensitive connection,

adding liquid to the other of said stages and pressurizing said other stage while so connected to activate said pressure-sensitive connection and without pressurizing said one stage,

and thereupon pressurizing said one stage through a lateral valve formed in the body of said one stage.

15. The method according to claim **14,**

further comprising loading with fuel and securing a third stage to said one of said two stages through a pressure-sensitive connection on said one stage and before pressurizing said one stage,

and thereupon pressurizing said third stage through a lateral valve opening formed in the body thereof.

16. The method of fuel loading and pressurizing two separable stages of a jet-propelled multistage missile comprising

adding liquid to a liquid chamber within one of said stages,

securing said stages together through a pressure-sensitive connection,

adding liquid to and pressurizing said other stage while so connected to activate said pressure-sensitive connection and without pressurizing said one stage,

and thereupon pressurizing said one stage throught the body of said one stage and other than through said other stage.

17. The method of connecting and pressurizing two separable linearly connectible units of a jet-propelled missile arrangement, comprising

securing said units together through a pressure-sensitive connection,

pressurizing the rearmost of said two units while so connected, to activate said pressure-sensitive connection and without pressurizing the other said unit,

and thereupon pressurizing the other said unit other than through said one unit.

18. The method according to claim **17,**

said units being sequential stages of a multi-stage missile.

19. The method of claim **17,**

said pressurizing of said other unit being effected through the body of said other unit and other than through said one unit.

20. The method of connecting and pressurizing two separable linearly connecting units of a jet-propelled missile arrangement, in which one of said units has a jet-propulsion fluid discharge orifice, comprising

securing said units together through a pressure-sensitive connection,

and pressurizing said one of said units through a valve spaced from said jet-propulsion fluid discharge orifice, to activate said pressure-sensitive connection.

21. The method according to claim **20,**

said one unit being the rearmost of said two units.

22. The method according to claim **20,**

said pressurizing being effected through a side wall valve.

23. The method according to claim **22,**

and self-sealing said valve after pressurizing therethrough.

24. The method according to claim **20,**

and self-sealing said valve after pressurizing therethrough.

25. The method according to claim **20,** in which the other said unit is pressurizable and has a jet propulsion fluid discharge orifice,

and thereupon pressurizing the other said unit other than through said one unit and other than through said jet propulsion fluid discharge orifice of said other unit.

26. The method according to claim **25,**

said pressurizing of said other unit being effected through a valve in said other unit and spaced from said jet propulsion fluid discharge orifice of said other unit.

27. The method of connecting and pressurizing two separable linearly connectible units of a jet-propelled missile arrangement, preparatory to launching, comprising

securing said units together through a pressure-sensitive connection,

pressurizing the rearmost of said two units while so connected, to activate said pressure-sensitive connection and without pressurizing the other said unit.

28. A self-propelled toy missile arrangement, comprising

a plurality of separable units,

one of said units being self-propelled and having a pressurizable fluid chamber and associated fluid discharge propulsion orifice,

said one of said units having an individually pressurizing enabling valve therein disposed other than at its fluid discharge orifice and for pressurizing said one unit while said units are assembled.

29. A multiunit toy missile arrrangement according to claim **13,**

two of said units having a respective selectively closable fluid discharge orifice and an associated respective pressurizable fluid chamber.

30. A multiunit toy missile arrangement according to claim **29,**

said two units being individually self-propellable,

said two respective fluid discharge orifices being selectively closable fluid discharge propulsion orifices.

31. A multiunit toy missile arrangement according to claim **13,**

at least two of said units being individually self-propellable and having selectively-closable fluid discharge propulsion orifices.

32. A self-propelled toy missile arrangement according to claim **28,**

said one unit having a pressurizable propulsion-discharge-fluid storage chamber in fluid connection between said pressurizing-enabling valve and said fluid discharge propulsion orifice,

said fluid discharge propulsion orifice connecting with said chamber and disposed in said one unit for pressure-responsive propulsive discharge therethrough under the residual internal pressure of pressurized fluid contained in said pressurized fluid storage chamber.

33. A self-propelled jet propulsion unit comprising

a spatially self-movable and -propelled jet propulsion body having a pressurizable propulsion-fluid storage chamber and a closable fluid jet propulsion discharge orifice connecting between said chamber and the outside atmosphere for propulsive discharge of pressurized fluid therefrom,

and a fluid pressurizing valve in said body in fluid pressurizing connection with said pressurizable fluid chamber and spaced from said jet propulsion discharge orifice and in fluid flow communication therewith through said pressurizable fluid propulsion storage chamber,

said self-propelled jet propulsion unit being self-propellable by force reaction to jet discharge of pressurizable fluid from said pressurizable fluid storage chamber and out through said jet propulsion discharge orifice in response to the internal fluid pressure of a quantity of pressurized fluid while in said pressurizable propulsion fluid storage chamber.

34. A self-propelled jet propulsion unit according to claim **33,**

said fluid pressurizing valve being a self-closing normally closed valve.

35. A self-propelled jet propulsion unit according to claim **33,**

said fluid pressurizing valve comprising a soft elastic material having a self-sealing normally self-closed bore.

36. A self-propelled jet propulsion unit according to claim **33,**

said body having an elongate shape, with said jet propulsion discharge orifice being disposed in one longitudinal end thereof,

said pressurizing valve being disposed with its outer end extending through an outer surface of said body spaced longitudinally from said fluid jet propulsion discharge orifice.

37. A self-propelled jet propulsion unit according to claim **36,**

said body having an inter-unit connector formed by an inflatably flexibly movable wall section spaced longitudinally from said fluid jet propulsion discharge orifice and said valve.

38. A multi-unit toy missile arrangement,

at least one unit thereof being individually self-propellable and having a respective pressurizable fluid chamber and associated fluid propulsion discharge orifice,

said one individually self-propellable unit having a body including a pressurizing-enabling element as a part thereof and being other than a fluid propulsion discharge orifice, for pressurizing said one unit from an external fluid pressure source through insertion of non-combustibly fluid-pressure-responsively-jet-dischargeable fluid under pressure into said one unit for subsequent non-combustibly pressure-responsive jet propulsion discharge through said fluid propulsion discharge orifice.

39. The method of connecting and pressurizing two separable,

connectible units of a jet-propelled missile arrangement, in which one of said units has a jet-propulsion fluid discharge orifice, comprising securing said units together through a pressure-sensitive connection, and pressurizing said one of said units having a jet propulsion fluid discharge orifice by injection of non-combustibly-pressure-responsively dischargeable jet propulsion fluid under pressure through a valve spaced from said jet propulsion fluid discharge orifice, to activate said pressure-sensitive connection,

said non-combustibly jet propulsion dischargeable fluid being subsequently non-combustibly dischargeable through said jet propulsion discharge orifice in response to the residual pressure within said one unit after fluid-pressure connecting-actuation of said pressure-sensitive connection as a function of the injected fluid pressure.

40. A self-propelled jet propulsion unit comprising

a spatially self-movable and -propelled jet propulsion body having a pressurizable propulsion-fluid storage chamber, with a closable fluid jet propulsion orifice connecting between said chamber and the outside atmosphere for propulsive discharge or pressurized fluid therefrom,

and a normally self-closed pressurized-fluid-containment fluid passage valve in said body at other than said fluid jet propulsion orifice for fluid passage connection between said pressurizable propulsion-fluid storage chamber and a zone exterior of said body when said normally self-closed valve is opened.

41. A unit according to claim **40**,

said valve comprising an opening formed in a wall of said body at other than said fluid jet propulsion orifice and which wall bounds a portion of said chamber, and a normally self-sealed seal member secured across said opening and being formed of soft sealing material relative to the less yielding fluid-pressure-containment material of said body.

42. A unit according to claim **41**,

said valve seal member comprising an insert of soft elastic sealing material disposed in secured relation at said opening.

43. A unit according to claim **39**,

said fluid passage valve being a fluid pressurizing valve comprising a soft elastic material having a self-sealing normally self-closed bore.

* * * * *